KSJ RESEARCH

Research Note Vol.3

Pricing an IPO in a 40x P/E Sector

An indicative price range for a hypothetical power-equipment maker — multiples, discounts and cycle risk from Korean and U.S. peers

Shinjae Kang

2026.10

© 2026 Shinjae Kang

Indicative range KRW 79,500–89,400: a 16–25% discount to the 43.0x P/E value is what covers cycle-peak risk

Three-step peer selection → applied multiple (P/E main, EV secondary) → value → discount → range → Monte Carlo cycle check

Key metrics [*1,*2,*3,*4]

Key metric	Value	p.
Applied P/E current · normalised · conservative	**43.0x · 29.1x · 20.1x**	5
3 Korean names aggregate P/E (mean · median)	40.1x (48.9x · 34.7x)	4
U.S. core peers' P/E (Vertiv, Eaton, GE Vernova)	45.9x	4
EV/EBIT (secondary, 2 Korean names)	37.4x	4
Target Co. NI to parent FY2025A (assumed)	KRW 30.8bn	2
Fair equity value (current multiple)	KRW 1,323bn	6
Value per share (post-IPO shares)	**KRW 105,801**	6
Indicative range (discount 16–25%)	**KRW 79,500–89,400**	6
Offer size (range low–top)	KRW 248bn–KRW 279bn	6
1-yr value per share P10 · P50 · P90	**71,481 · 89,375 · 110,845**	7
1-yr loss probability at the top of the range	50%	7

Source: *1 FSS DART annual reports (consolidated) *2 Korea Exchange (KRX) daily trading data *3 SEC EDGAR 10-K XBRL (companyfacts), cover-page shares *4 KSJ Research
Note: KSJ Research. Target Co. is hypothetical; definitions and sources in footnotes and A2. Judgements are the author's own

43.0x

Applied P/E (Korea and U.S. core average, as of)

43.0x is an earnings-made multiple, but a peak one — normalised 29.1x, conservative 20.1x

- Normalised 29.1x: log midpoint of early-2023 20.1x and today (half the re-rating reverses)
- Vol.1: 61% of the same names' gain was earnings — the multiple itself signals a peak
- LS ELECTRIC at 86.1x exceeds 2x the median — kept under aggregate weighting (mean would be 48.9x)

KRW 105,801

Value per share (current multiple, post-IPO shares)

Value KRW 105,801 → indicative range KRW 79,500–89,400 (P25–P50 of 1-yr value)

- NI KRW 30.8bn × 43.0x = equity KRW 1,323bn ÷ 12.5mn shares post-IPO
- Offer 3.12mn shares (80% primary, 20% secondary) → KRW 248bn–KRW 279bn
- 10–40% discount scenarios: KRW 63,480–KRW 95,221
- EV/EBIT cross-check: KRW 138,587 per share (+31% vs P/E)

16–25%

Required discount (1-yr value at P50 · P25)

The price of cycle-peak risk: a 16–25% discount to the current-multiple value — the upper half of the 10–40% grid

- 1-yr value P10 71,481 · P50 89,375 · P90 110,845 (KRW; 10,000 runs, seed 42)
- At a 10% discount the 1-yr loss probability is 64%; at 40%, 2%
- Protecting the 10th percentile needs 32%

© 2026 Shinjae Kang · KSJ Research · Research Note Vol.3 · 2026.10 · For education and research only, not investment advice · Target Co. is hypothetical

Contents

Start → assumptions → peers → multiple → scenarios → price → Monte Carlo → implications: each section's conclusion

I **Starting point** 1
P/E 13.5x → 40.1x, 61% of the gain was earnings — earnings that were developers' costs

II **Target Co.** 2
Target Co. inputs are Korean peers' DART aggregates — growth 18.6%, operating margin 14.4%, net margin 10.3%

III **Peers** 3
3 Korean + 3 U.S. core peers, 2 generators as reference — Doosan Enerbility and Schneider excluded

IV **Applied multiple** 4
Applied P/E 43.0x — Korea aggregate 40.1x, mean 48.9x, median 34.7x; EV secondary

V **Multiple scenarios** 5
Current 43.0x · normalised 29.1x · conservative 20.1x — conservative values the same earnings at 47% of today

VI **Offer price** 6
Value KRW 105,801 per share → indicative range KRW 79,500–89,400 — offer KRW 248bn–KRW 279bn

VII **Monte Carlo** 7
1-yr P50 = 84% of value — required discount 16%–25%

VIII **Implications** 8
In a 43x P/E sector, price the IPO at a 16%–25% discount, not at the peak multiple

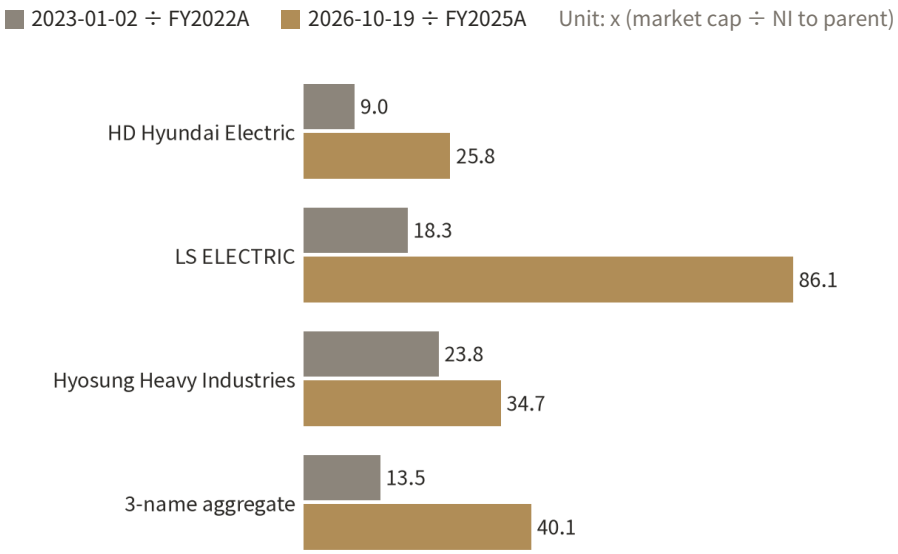
App. **Appendix** A1–A5
A1 Q&A · A2 Assumptions and sources · A3 Data notes · A4 Methodology · A5 Glossary

© 2026 Shinjae Kang · KSJ Research · Research Note Vol.3 · 2026.10 · For education and research only, not investment advice · Target Co. is hypothetical

Starting point: P/E 13.5x → 40.1x, 61% of the gain was earnings — earnings that were developers' costs

Vol.1 earnings-led re-rating + Vol.2 equipment pricing power → what range for a hypothetical unlisted company at this multiple?

3-name aggregate P/E 13.5x (early 2023) → 40.1x (as of) — 61% of the gain was earnings [*1,*2,*3]



Source: *1 Korea Exchange (KRX) daily trading data *2 FSS DART annual reports (consolidated) *3 KSJ Research Vol.1 (Oct 2026)
Note: Vol.1 log split: Δln(cap) = Δln(NI) + Δln(P/E). Doosan Enerbility excluded (FY2022A net loss)

Vol.2: peer margin 5.2% → 14.4% — if all from price, equipment prices +10.9% (upper bound), the developer's cost [*1,*2]

Company	OPM FY2022A	OPM FY2025A	Implied price rise
HD Hyundai Electric	6.3%	24.4%	23.9%
LS ELECTRIC	5.6%	8.6%	3.3%
Hyosung Heavy Industries	4.1%	12.5%	9.6%
3-name total	5.2%	14.4%	**10.9%**

Source: *1 FSS DART annual reports (consolidated) *2 KSJ Research Vol.2 (Oct 2026)
Note: Implied rise = (1 − OPM start) ÷ (1 − OPM end) − 1: an upper bound with unit cost unchanged (not causal). That profit is today's P/E denominator

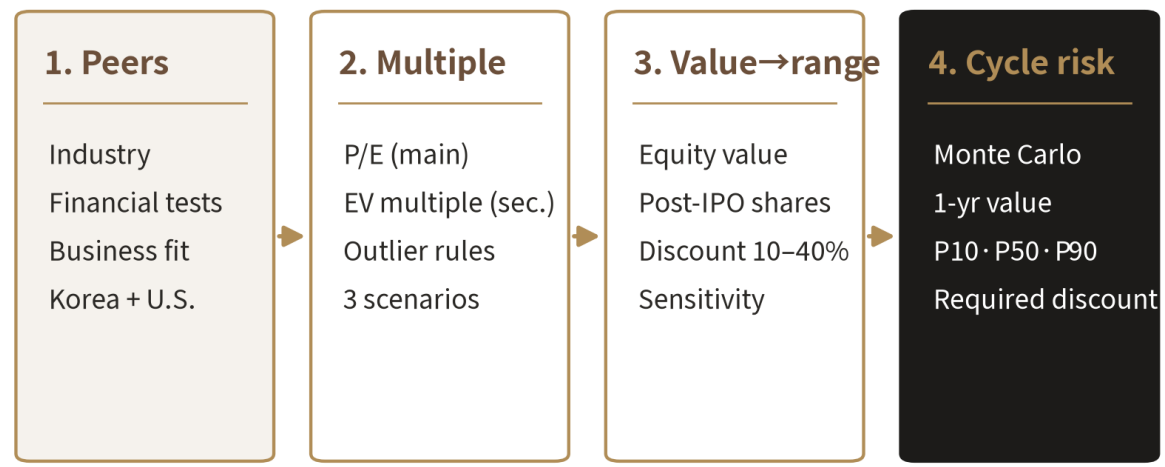
Target Co.: a hypothetical unlisted power-equipment maker

Question: can an earnings-made 40x P/E be applied to an unlisted company's IPO price?

- Hypothetical (not a real company): a Korean mid-size maker of switchgear, UPS and data-centre power distribution
- FY2025A revenue KRW 300bn (design) — growth, margins and net debt from Korean peers' DART data
- 10mn shares pre-IPO · offer 25% of post-IPO shares (80% primary) → 12.5mn shares
- As of 2026-10-20 — caps at the prior trading day, earnings FY2025A (latest fiscal year)

Note: Assumptions and their status on p2 and A2

Korean IPO practice: value → discount → indicative range [*1]



Source: *1 KSJ Research
Note: A generic structure, not a copy of any filing; final pricing (book-building) is out of scope

© 2026 Shinjae Kang · KSJ Research · Research Note Vol.3 · 2026.10 · For education and research only, not investment advice · Target Co. is hypothetical

Target Co. inputs are Korean peers' DART aggregates — growth 18.6%, operating margin 14.4%, net margin 10.3%

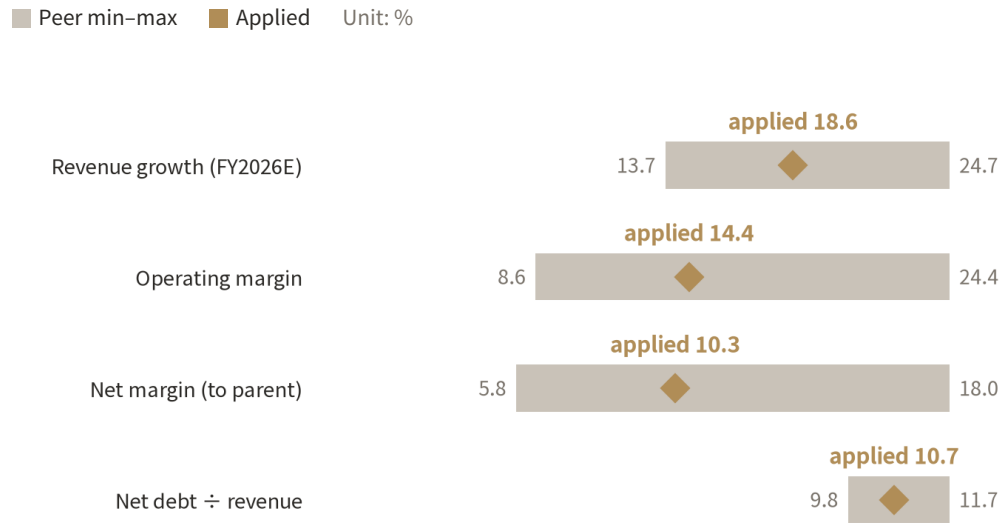
Revenue KRW 300bn and 10mn shares pre-IPO are design values (price unit); the rest are peer aggregates within the peer range

4 of 9 Target Co. inputs come from Korean peers' DART data — ranges are the min–max of 3 names [*1,*2,*3]

Item	Applied	Peer range	Status
Revenue FY2025A	3,000	—	Design
Revenue growth (FY2026E)	18.6%	13.7%–24.7%	Data
Operating margin	14.4%	8.6%–24.4%	Data
Net margin (to parent)	10.3%	5.8%–18.0%	Data
Net debt ÷ revenue	10.7%	9.8%–11.7%	Data
D&A ÷ revenue	—	—	Input needed
Pre-IPO shares	10,000,000	—	Design
Offer size (% of post-IPO shares)	25.0%	—	Verify
Primary share (rest secondary)	80.0%	—	Design

Source: *1 FSS DART annual reports (consolidated) *2 Framework Act on SMEs Enforcement Decree Annex 1; KRX listing rules (free-float requirement) *3 KSJ Research
Note: Applied = 3-name combined (growth = combined revenue CAGR FY2022A→FY2025A). Status: data = DART, design = hypothetical setting, verify = public rule pending check

Applied values (◆) sit inside the peer range — margins from LS ELECTRIC 8.6% to HD Hyundai Electric 24.4% [*1]



Source: *1 FSS DART annual reports (consolidated)
Note: Net-debt ratio uses names with debt data (HD Hyundai Electric: debt not collected)

FY2025A NI to parent KRW 30.8bn — the earnings the price is built on [*1,*2]

Unit: KRW 100mn → bn, KRW

Item	FY2025A	FY2026E (growth applied)
Revenue	KRW 300bn	KRW 356bn
Operating income	KRW 43.3bn	KRW 51.4bn
NI to parent	**KRW 30.8bn**	KRW 36.5bn
EBIT	KRW 43.3bn	—
Net debt	KRW 32.0bn	—
EPS (pre-IPO shares)	KRW 3,075	KRW 3,649

Source: *1 FSS DART annual reports (consolidated) *2 KSJ Research
Note: FY2026E = FY2025A × (1 + 18.6%), margins fixed. The price applies peers' P/E on the same basis (FY2025A)

How inputs move the result: scale is irrelevant; margins and multiples matter

- Revenue scale and share count only set the price unit: value = earnings × multiple; the discount conclusion does not depend on scale
- 1pp of net margin = KRW 3.0bn NI = KRW 129bn of value at the current multiple
- Margins are cyclical too: 3-name net margin FY2022A 3.1% → FY2025A 10.3%, operating 5.2% → 14.4%
- At FY2022A margins, NI on the same revenue would be 31% of today's — multiple and earnings may both be at a peak

Note: The margin cycle is not in the Monte Carlo; see limitations (p8) and Q&A (A1)

Peers: 3 Korean + 3 U.S. core equipment makers; 2 generators as reference — Doosan Enerbility at 503x P/E and Schneider Electric (not an SEC filer) excluded

Prospectus-style three steps: industry (power infrastructure) → financial tests (profit, multiple, filer) → business fit (distribution, UPS, data-centre power)

10 candidates → 6 applied — industry → financial tests → business fit [*1,*2,*3]

Company	1. Industry	2. Financial tests	3. Business fit	Result
HD Hyundai Electric	Power equipment	Profitable · P/E 25.8x	Transformers, HV equipment (medium)	Applied
LS ELECTRIC	Power equipment	Profitable · P/E 86.1x	Switchgear, breakers (high)	Applied
Hyosung Heavy Industries	Power equipment	Profitable · P/E 34.7x	Transformers, breakers (heavy industries) (medium)	Applied
Doosan Enerbility	Generation equip.	P/E 503x > 100x	Nuclear and gas-turbine generation equipment (low)	Excluded
Vertiv	Power equipment	Profitable · P/E 59.5x	Data-centre UPS, power distribution, thermal (high)	Applied
Eaton	Power equipment	Profitable · P/E 37.2x	Power distribution, UPS, power management (electrical) (high)	Applied
GE Vernova	Power equipment	Profitable · P/E 49.5x	Gas turbines, wind, grid equipment (Electrification) (high)	Applied
Schneider Electric	Power equipment	Not an SEC filer	Distribution, power management, automation (similar business)	Excluded
Constellation Energy	Power generation	Profitable · P/E 40.2x	Nuclear-led generator (generation)	Reference
Vistra	Power generation	Profitable · P/E 53.2x	Generator and retailer (generation)	Reference

Source: *1 FSS DART annual reports (consolidated) *2 Korea Exchange (KRX) daily trading data *3 SEC EDGAR 10-K XBRL (companyfacts), cover-page shares
Note: Financial tests: filed financials, FY2025A profit, P/E ≤ 100x, listed entity = filer (SEC). Business notes summarise annual-report segments (author)

Equipment makers and generators are different businesses — net debt ÷ revenue 2% vs 58% [*1]

Company	Group	Rev. CAGR 23→25	OPM	Net margin	Net debt ÷ rev.
Vertiv	Core · equipment	22.1%	17.9%	13.0%	12%
Eaton	Core · equipment	8.8%	—	14.9%	34%
GE Vernova	Core · equipment	7.0%	3.6%	12.8%	-23%
Constellation Energy	Ref. · generation	4.3%	13.6%	10.2%	24%
Vistra	Ref. · generation	12.9%	10.8%	5.4%	103%

Source: *1 SEC EDGAR 10-K XBRL (companyfacts), cover-page shares
Note: SEC 10-K FY2025A (USD); revenue CAGR FY2023A→FY2025A. Equipment = orders and production, margins from pricing power / generation = power sales and PPAs with heavy assets and debt, so P/E and EV behave differently → reference only. Eaton reports no operating-income tag (—)

Market cap = close × SEC shares — 0 outside ±2.0% of yfinance shares [*1,*2,*3]

Company	Close USD	SEC shares (mn)	yfinance (mn)	Diff.	Market cap	In KRW
Vertiv	206.08	385.0	385.0	+0.0%	USD 79,338mn	KRW 105,687bn
Eaton	391.50	388.4	388.4	+0.0%	USD 152,059mn	KRW 202,557bn
GE Vernova	908.44	266.3	266.3	+0.0%	USD 241,948mn	KRW 322,299bn
Constellation Energy	263.19	354.3	354.3	+0.0%	USD 93,250mn	KRW 124,219bn
Vistra	149.70	335.6	335.6	+0.0%	USD 50,245mn	KRW 66,931bn

Source: *1 yfinance (Yahoo Finance, secondary; cross-checked with SEC shares) *2 SEC EDGAR 10-K XBRL (companyfacts), cover-page shares *3 Bank of Korea ECOS USD/KRW
Note: yfinance is secondary — SEC cover shares are used; the difference is checked only. KRW = USD mn × 1,332.1 ÷ 100 (ECOS 2026-10-20)

Exclusions and checks: distorted multiple, disclosure scope, filer identity

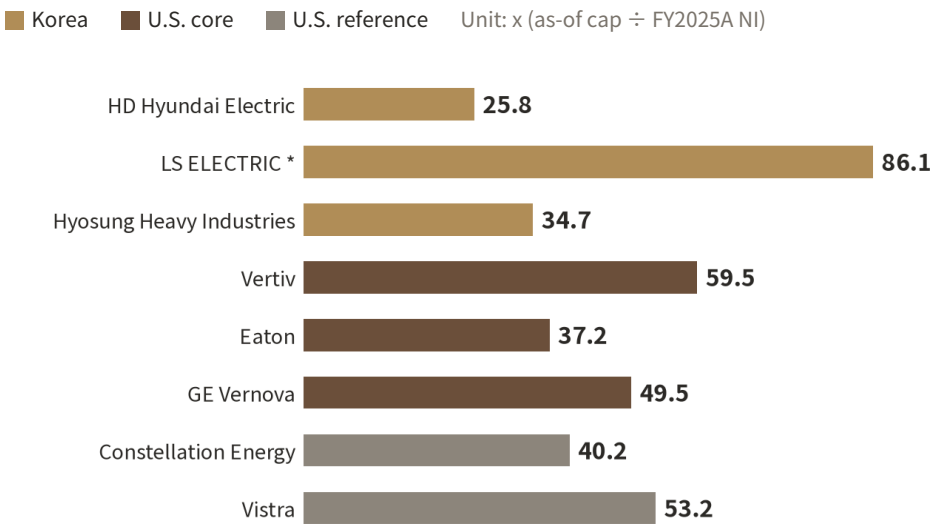
- Doosan Enerbility: P/E 503x (denominator distortion from FY2025A NI of KRW 84.8bn) and low fit (generation equipment). Also excluded from Vol.1's split (FY2022A loss)
- Schneider Electric: Not an SEC filer (Euronext Paris) — manual-entry template only; excluded this volume
- Vistra: automatic entity check WARN → OK after checking the 10-K cover registrant (2026-10-22)
- GE Vernova: no market cap on the first 2023 trading day → excluded from the conservative multiple

Note: Outlier rules in the p4 table

Applied P/E 43.0x: Korea aggregate 40.1x (mean 48.9x, median 34.7x) + U.S. core 45.9x

P/E (main) · EV/EBIT (secondary) · outlier rules R1–R5 — Korea, U.S. and combined computed separately

Current P/E: Korea 25.8x–86.1x · U.S. core 37.2x–59.5x — LS ELECTRIC 86.1x exceeds 2x the median (*, kept) *1,*2,*3



Source: *1 Korea Exchange (KRX) daily trading data *2 FSS DART annual reports (consolidated) *3 SEC EDGAR 10-K XBRL (companyfacts), cover-page shares
Note: Korea = KRX 2026-10-19 cap ÷ DART FY2025A NI to parent. Excluded names not shown (p3)

Applied P/E 43.0x — aggregate within groups, average across groups *1,*2,*3

Group	n	Aggregate	Mean	Median	Applied
3 Korean names	3	40.1x	48.9x	34.7x	40.1x
U.S. core (equipment)	3	45.9x	48.8x	49.5x	45.9x
U.S. reference (generation)	2	44.0x	46.7x	46.7x	compare only
Applied (Korea and U.S. core average)					**43.0x**

Source: *1 Korea Exchange (KRX) daily trading data *2 FSS DART annual reports (consolidated) *3 SEC EDGAR 10-K XBRL (companyfacts), cover-page shares
Note: Aggregate = Σcap ÷ ΣNI. With 3–5 peers a simple mean is driven by one name, so the aggregate leads, mean and median shown. Korea and U.S. are averaged as groups (very different sizes)

Outlier rules: drop losses and extreme multiples; flag, but keep, high ones *1

Rule	Test	Treatment	Hit
R1	NI to parent ≤ 0	Drop P/E	—
R2	P/E > 100x	Drop (denominator distortion)	Doosan Enerbility
R3	Listed entity ≠ filer (SEC)	Drop	—
R4	> 2x group median	Flag, keep (aggregate)	LS ELECTRIC
R5	EBITDA ≤ 0, EV multiple > 60x, or no net debt	Drop EV multiple	GE Vernova, HD Hyundai Electric

Source: *1 KSJ Research
Note: Rules set before looking at the multiples (author's judgment); the same rules apply to the early-2023 point

EV/EBIT 40.7x (secondary) — applied to Target, value is +31% vs P/E *1,*2,*3

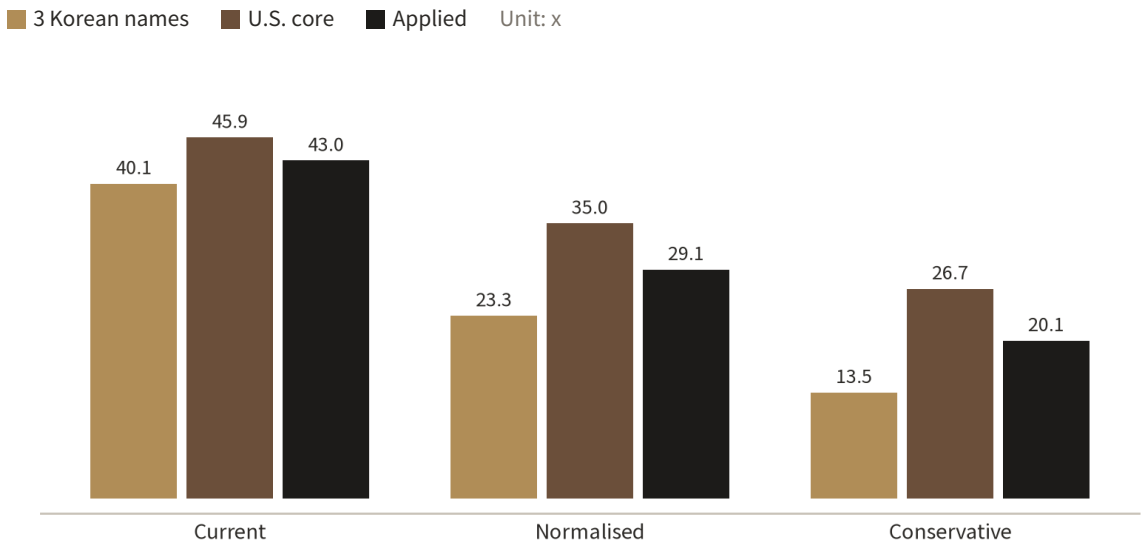
Company	EV	EBIT	Multiple	Treatment
HD Hyundai Electric	—	KRW 995bn	—	No net debt / cap
LS ELECTRIC	KRW 25,270bn	KRW 426bn	59.3x	Kept
Hyosung Heavy Industries	KRW 18,638bn	KRW 747bn	25.0x	Kept
Vertiv	USD 80,523mn	USD 1,830mn	44.0x	Kept
Eaton	USD 161,332mn	—	—	No EBIT
GE Vernova	USD 233,163mn	USD 1,388mn	168.0x	R5

Source: *1 FSS DART annual reports (consolidated) *2 Korea Exchange (KRX) daily trading data *3 SEC EDGAR 10-K XBRL (companyfacts), cover-page shares
Note: D&A not collected → EBIT stands in for EBITDA (EV/EBIT ≥ EV/EBITDA), applied to Target's EBIT for consistency. EV = cap + net debt (ex leases). Aggregate n = 2

Scenarios: current 43.0x · normalised 29.1x · conservative 20.1x — conservative values the same earnings at 47% of today

Conservative = observed early-2023 P/E (same peers, same rules); normalised = log midpoint — outlier rules applied at both points

Current 43.0x → normalised 29.1x → conservative 20.1x: equity value falls to 68% and 47% *1,*2,*3



Source: *1 Korea Exchange (KRX) daily trading data *2 FSS DART annual reports (consolidated) *3 SEC EDGAR 10-K XBRL (companyfacts), cover-page shares
Note: Scenarios are a range, not probabilities; the Monte Carlo (p7) uses a triangular distribution from conservative to current

All three are data: today and early 2023 observed; normalised is their log midpoint *1,*2,*3

Scenario	Definition	3 Korean names	U.S. core	Applied
Current	As-of (2026-10-19) cap ÷ FY2025A	40.1x	45.9x	**43.0x**
Normalised	√(conservative × current): 50% log reversal	23.3x	35.0x	**29.1x**
Conservative	First 2023 trading day (2023-01-02) cap ÷ FY2022A	13.5x	26.7x	**20.1x**

Source: *1 Korea Exchange (KRX) daily trading data *2 FSS DART annual reports (consolidated) *3 SEC EDGAR 10-K XBRL (companyfacts), cover-page shares
Note: Applied = simple average of Korea and U.S. core group multiples

By company: Korea early-2023 P/E 9.0x–23.8x → today *1,*2,*3

Company	Cap 2023-01-02	NI FY2022A	P/E early-23	P/E now	Change
HD Hyundai Electric	KRW 1,458bn	KRW 162bn	9.0x	25.8x	2.9x
LS ELECTRIC	KRW 1,653bn	KRW 90.3bn	18.3x	86.1x	4.7x
Hyosung Heavy Industries	KRW 693bn	KRW 29.1bn	23.8x	34.7x	1.5x
Doosan Enerbility	KRW 9,734bn	KRW -772bn	R1	503.3x	—
Vertiv	USD 5,018mn	USD 77mn	65.5x	59.5x	0.9x
Eaton	USD 62,737mn	USD 2,462mn	25.5x	37.2x	1.5x
GE Vernova	—	USD -2,736mn	R1	49.5x	—

Source: *1 Korea Exchange (KRX) daily trading data *2 FSS DART annual reports (consolidated) *3 SEC EDGAR 10-K XBRL (companyfacts), cover-page shares
Note: R1 = FY2022A net loss (Doosan Enerbility) → also excluded at the conservative point. U.S. peers after first-2023 closes and shares are collected

Why a log midpoint: the same scale as Vol.1's split

- Vol.1 split the gain in logs: earnings 61%, multiple 39%
- Normalised = 50% of the re-rating reverses on the same log scale → 29.1x
- An arithmetic midpoint would be 31.5x — tilted toward the high multiple
- Equity value: current KRW 1,323bn · normalised KRW 896bn · conservative KRW 618bn

Note: The 50% reversal is the author's judgment; the Monte Carlo spans conservative to current

Value KRW 105,801 per share → indicative range KRW 79,500–89,400 (16–25% discount) — offer KRW 248bn–KRW 279bn

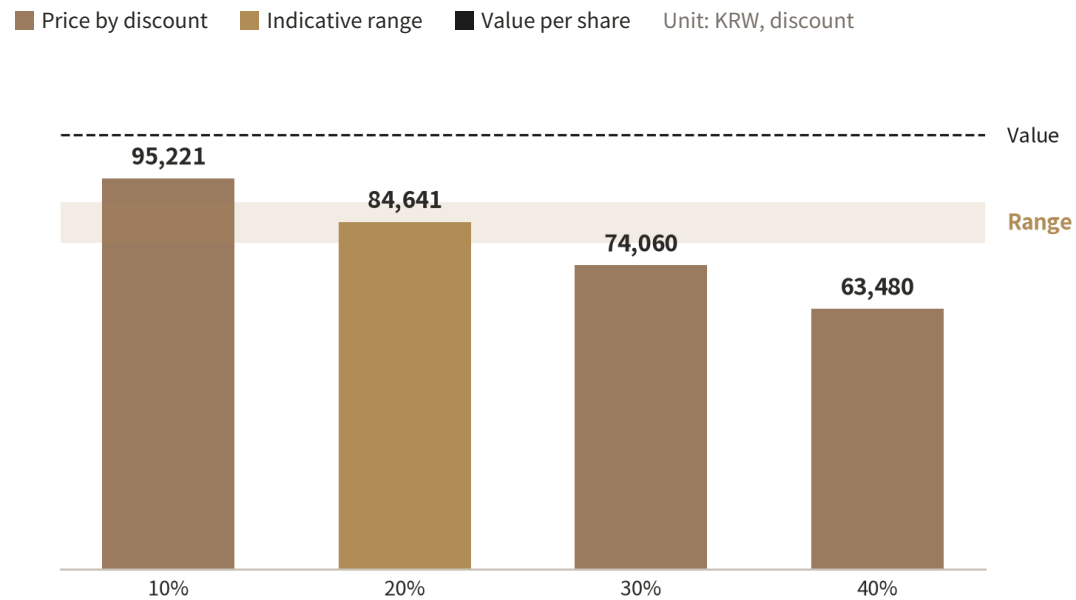
Equity value → post-IPO shares (primary, secondary) → value per share → discount scenarios (10–40%) → range · sensitivity (multiple × discount)

Value KRW 105,801 = KRW 30.8bn × 43.0x ÷ 12.5mn shares → range KRW 79,500–89,400 *1

	Step	Value
①	Earnings: Target Co. FY2025A NI to parent	KRW 30.8bn
②	× applied P/E (current)	43.0x
③	= Fair equity value	KRW 1,323bn
④	÷ post-IPO shares (10mn shares + 2.5mn shares new)	12.5mn shares
⑤	= Value per share	**KRW 105,801**
⑥	Discount (P25–P50 of 1-yr value)	16–25%
⑦	= Indicative range (rounded to KRW 100)	**KRW 79,500–89,400**

Source: *1 KSJ Research
Note: Value per share uses post-IPO shares (new-share proceeds not added — conservative). Range = rounded quantiles of 1-yr value (p7)

10–40% discount: KRW 63,480–KRW 95,221 — the range sits at 16–25% *1



Source: *1 KSJ Research
Note: Discount = 1 − price ÷ value per share (current multiple)

Sensitivity: 5x of P/E = KRW 12,302 per share (before discount) — cells inside the range in gold *1

Unit: KRW (rows = applied P/E, columns = discount)

Applied P/E	10%	20%	30%	40%
Current 43.0x	**95,221**	**84,641**	**74,060**	**63,480**
Normalised 29.1x	**64,510**	**57,342**	**50,175**	**43,007**
Conservative 20.1x	**44,488**	**39,545**	**34,602**	**29,659**
15x	33,214	29,524	25,833	22,143
20x	44,286	39,365	34,445	29,524
25x	55,357	49,207	43,056	36,905
30x	66,429	59,048	51,667	44,286
35x	77,500	68,889	60,278	51,667
40x	**88,572**	78,731	68,889	59,048
45x	99,643	**88,572**	77,500	66,429

Discount

Source: *1 KSJ Research
Note: Top three rows = scenarios (p5), below = 5x grid. Cells above the range in warm grey

Offer 3.12mn shares (80% primary) · KRW 248bn–KRW 279bn — existing holders' 75% is the overhang *1,*2

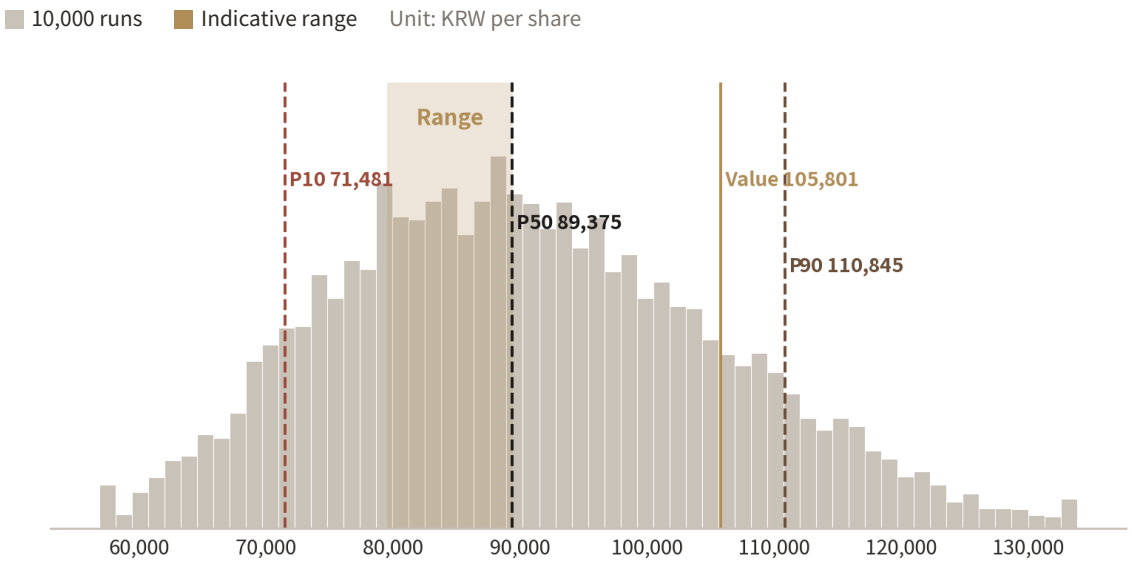
Item	Shares	Amount (range low–top)
Pre-IPO shares	10,000,000	—
Offer shares (25% of post-IPO)	3,125,000	KRW 248bn–KRW 279bn
Primary (to company)	2,500,000	KRW 199bn–KRW 224bn
Secondary (to sellers)	625,000	KRW 49.7bn–KRW 55.9bn
Post-IPO shares	12,500,000	Cap KRW 994bn–KRW 1,118bn
Existing holders' stake (overhang)	9,375,000	75.0%
Dilution from new shares		20.0%

Source: *1 Framework Act on SMEs Enforcement Decree Annex 1; KRX listing rules (free-float requirement) *2 KSJ Research
Note: Offer size at the listing rules' free-float level (to verify); primary share is hypothetical. Lock-up not set → shares only

1-yr P50 value KRW 89,375 = 84% of today's value — covering cycle-peak risk takes 16% (P50) to 25% (P25)

10,000 runs · seed 42 · one year of earnings growth × 1-yr multiple (conservative to current) — where the range sits and loss odds

1-yr value P10 71,481 · P50 89,375 · P90 110,845 — the KRW 105,801 value sits at P84 *1,*2,*3



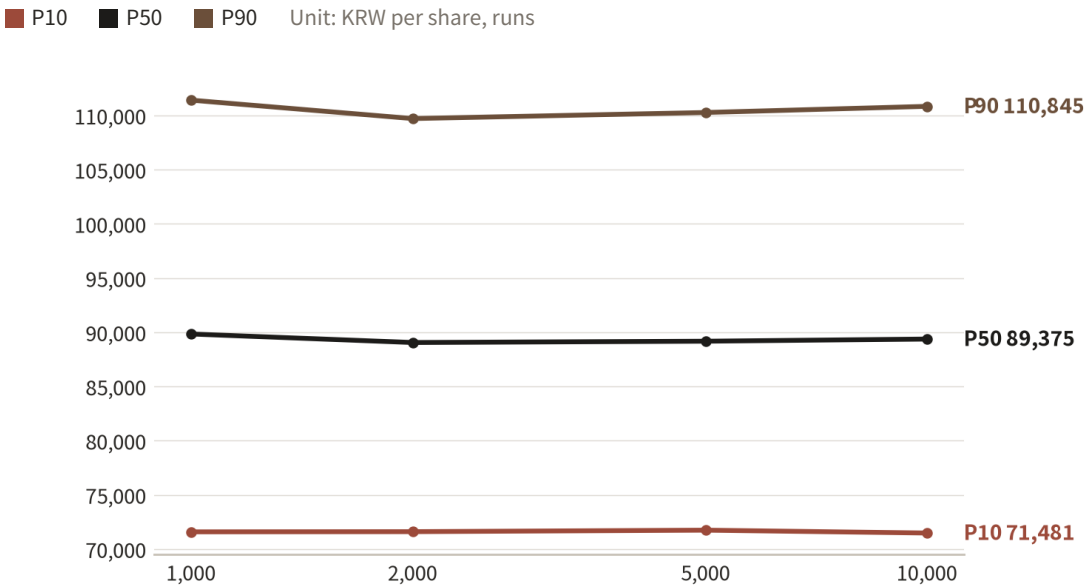
Source: *1 FSS DART annual reports (consolidated) *2 Korea Exchange (KRX) daily trading data *3 KSJ Research
Note: 1-yr value = FY2025A NI × (1 + growth) × 1-yr-ahead P/E ÷ post-IPO shares; margins fixed

At 10% the 1-yr loss probability is 64%, at 40% 2% — a 50% loss probability needs 16% *1

Discount / basis	Price	Loss prob. · needed discount
10%	KRW 95,221	64%
20%	KRW 84,641	38%
30%	KRW 74,060	14%
40%	KRW 63,480	2%
Range top	KRW 89,400	50%
Range low	KRW 79,500	25%
Needed (P50)	KRW 89,375	**16%**
Needed (P25)	KRW 79,475	**25%**
Needed (P10)	KRW 71,481	32%

Source: *1 KSJ Research
Note: Loss probability = P(1-yr value < price). Bottom three rows = discount needed to price at that quantile (vs value)

Convergence: P50 moves 0.22% from 5,000 to 10,000 runs (seed 42) *1



Source: *1 KSJ Research
Note: First n draws of one seeded run. Converged if the last P50 change ≤ 1% → converged

Only two inputs move: earnings growth (9 observations) and the 1-yr multiple (scenario triangle) *1,*2,*3

Input	Distribution / value	Basis
Earnings growth (1 yr)	ln(1+g) ~ N(0.175, 0.063)	9 Korean peer revenue-growth observations
Growth median · 90% range	19.2% · 7.5~32.1%	Log-normal; negative growth possible
1-yr-ahead P/E	Tri(20.1, 29.1, 43.0)	Min conservative · mode normalised · max current (neutral)
Margins · shares	Net margin 10.3% fixed · 12.5mn shares	Margin cycle excluded (limitation)
Runs · seed	10,000 · 42	Reproducible
Range basis	P25 · P50	Low 25% · top 50% loss probability (judgment)

Source: *1 FSS DART annual reports (consolidated) *2 Korea Exchange (KRX) daily trading data *3 KSJ Research
Note: Growth and multiple are drawn independently; in reality slower growth and de-rating may coincide, fattening the downside

© 2026 Shinjae Kang · KSJ Research · Research Note Vol.3 · 2026.10 · For education and research only, not investment advice · Target Co. is hypothetical

Implication: in a 43x P/E sector, price the IPO at a 16%–25% discount, not at the peak multiple

ECM checklist · limitations · next: Vol.4 (data-centre power supply) · conclusion (cycle-peak risk and the discount)

ECM checklist: which multiple, and how much discount, comes first [*1]

Item	This note	Check in a real deal	Pages
Peer set	3 Korean + 3 U.S. core (after caps), 2 reference	Add distribution/UPS-heavy peers; Schneider from its annual report	3
Applied multiple	Aggregate P/E 43.0x (mean 48.9x, median 34.7x)	Whether the R4-flagged name stays moves the mean a lot	4
Earnings basis	FY2025A (latest year) — same pairing as peers	Update to the last four quarters (LTM)	2, 4
Discount	**16–25% (P25–P50 of 1-yr value)**	Compare with recent Korean IPO discounts	6, 7
Secondary sale · overhang	Secondary 20% · existing holders 75%	Lock-up terms and release dates; reason for selling	6
Cycle risk	1-yr P50 = 84% of value	Check backlog and margin trends for a peak	5, 7

Source: *1 KSJ Research
Note: Checklist for the hypothetical company; figures from the body

Limitations of this note

- Target Co. is hypothetical: size, shares and structure are design values; growth and margins are Korean peer aggregates
- D&A not collected → EV/EBIT instead of EV/EBITDA (secondary)
- Earnings are FY2025A (closed ~10 months ago); LTM would change both multiple and earnings
- The Monte Carlo varies growth and multiple only (independent), margins fixed — downside may be understated
- Range quantiles (P25–P50), the 50% reversal and outlier thresholds are judgments — shown with sensitivities

Note: Updated with the same code as data arrive

Next in Vol.4: powering data centres — generation assets and PPAs

After equipment, the power itself: who sells electricity to data centres, and under what contracts?

- Valuing generators (Constellation, Vistra — this volume's reference peers) and PPA structures
- How PPA price and tenor feed Vol.2's power cost and debt capacity
- Vol.5 finale: an AI energy asset investment map — demand, equipment, PF, IPO and generation on one page

Note: Timing and method of Vol.4–5 may change

Conclusion: how much discount for cycle-peak risk?

- Pricing at the current-multiple value of KRW 105,801 carries a 84% 1-yr loss probability
- Breaking even at the median takes 16%; protecting the 25th percentile 25%
- So cycle-peak risk is worth a 16%–25% discount — the upper half of the 10–40% grid
- To narrow the discount, prove earnings durability (backlog, long-term contracts), not the multiple

Note: From the Monte Carlo (p7)

© 2026 Shinjae Kang · KSJ Research · Research Note Vol.3 · 2026.10 · For education and research only, not investment advice · Target Co. is hypothetical

Q&A: questions on the multiple, the discount and the structure

Questions likely in deal work (valuation, prospectus, investor meetings), with page references

Eight questions an ECM desk would ask [*1,*2,*3,*4]

Question	Answer (from the body)	p.
Shouldn't LS ELECTRIC at 86.1x be dropped?	Flagged (R4, > 2x median), but the aggregate is earnings-weighted: 40.1x with vs 29.5x without. A simple mean would be 48.9x	4
Why is Doosan Enerbility excluded?	P/E 503x (NI KRW 84.8bn) is a denominator distortion (R2), and its nuclear/gas-turbine mix fits poorly	3
Why P/E first and EV/EBITDA second?	IPO prices are set on equity value (NI to parent), and net-debt/D&A data are incomplete. The EV/EBIT cross-check is +31% vs P/E	4
Does adding U.S. peers change the answer?	U.S. core at 45.9x is averaged with Korea for 43.0x	3, 4
Why are Constellation and Vistra only references?	They sell power: profits depend on prices and PPAs, and heavy assets and debt make P/E and EV behave unlike equipment makers (p3 net debt ÷ revenue)	3
Isn't using peak peer margins aggressive?	Yes. 3-name net margin went FY2022A 3.1% → FY2025A 10.3%; at FY2022A margins the value would be 31% — the discount does not cover the margin cycle (limitation)	2, 8
Isn't a 16–25% discount excessive?	At 10% and 20% the 1-yr loss probabilities are 64% and 38%. A smaller discount needs evidence the multiple holds (backlog, long-term contracts)	7
How should the secondary sale and overhang be read?	20% of the offer is secondary (5.0% of post-IPO). Existing holders' 75% is overhang once lock-ups end — disclosed and checked separately from the discount	6

Source: *1 FSS DART annual reports (consolidated) *2 Korea Exchange (KRX) daily trading data *3 SEC EDGAR 10-K XBRL (companyfacts), cover-page shares *4 KSJ Research
Note: All figures in the answers come from the same calculations as the body

© 2026 Shinjae Kang · KSJ Research · Research Note Vol.3 · 2026.10 · For education and research only, not investment advice · Target Co. is hypothetical

Assumptions and sources: data, sourced, judgment and design kept apart

Target Co., deal structure and Monte Carlo inputs — value, peer range, basis, status

18 inputs: data 6 · sourced 0 · verify 2 · design 3 · judgment 2 · needed 4 [*1,*2,*3]

Item	Value (range)	Unit	Basis / source	Status
Revenue FY2025A	3,000	KRW 100mn	Hypothetical company size, FY2025A — above the mid-size threshold; sets the price unit only	Design
Mid-size revenue threshold (ref.)	1,500	KRW 100mn	Framework Act on SMEs, Enforcement Decree Annex 1 (electrical equipment: KRW 150bn average revenue)	Verify
Revenue growth (FY2026E)	18.6% (13.7–24.7)	%	DART, 3-name combined revenue CAGR FY2022A→FY2025A (range = min–max)	Data
Operating margin	14.4% (8.6–24.4)	%	DART, 3-name combined operating margin FY2025A (range = min–max)	Data
Net margin (to parent)	10.3% (5.8–18.0)	%	DART, 3-name combined net margin to parent FY2025A (range = min–max)	Data
Net debt ÷ revenue	10.7% (9.8–11.7)	%	DART, net debt ÷ revenue FY2025A (names with debt data; range = min–max)	Data
D&A ÷ revenue	—	%	Not collected → EV/EBIT as proxy	Input needed
Pre-IPO shares	10,000,000	shares	Hypothetical pre-IPO share count — changes the price level only	Design
Offer size (% of post-IPO shares)	25.0%	%	Offer as % of post-IPO shares — KRX listing rules' free-float requirement (e.g. 25% public holders)	Verify
Primary share (rest secondary)	80.0%	%	Primary share of the offer (rest secondary sale) — hypothetical deal design	Design
Lock-up (major holders)	—	months	Not set → overhang shown in shares only	Input needed
Band discount (low)	—	%	Not sourced → band from Monte Carlo quantiles	Input needed
Band discount (high)	—	%	Not sourced → band from Monte Carlo quantiles	Input needed
Growth mean (log)	17.5% (7.3–25.0)	log	DART, 9 annual revenue growth observations FY2023A–FY2025A (log mean)	Data
Growth s.d. (log)	6.28%	log	Same 9 observations (log standard deviation)	Data
1-yr-ahead P/E distribution	Triangular	-	Triangular: min conservative, mode normalised, max current — neutral shape without probability evidence	Neutral
Band low quantile	25.0%	%	Band low = 25th percentile of 1-yr value (25% loss probability at the low end)	Judgment
Band high quantile	50.0%	%	Band top = median 1-yr value (50% loss probability at the top)	Judgment

Source: *1 FSS DART annual reports (consolidated) *2 Framework Act on SMEs Enforcement Decree Annex 1; KRX listing rules (free-float requirement) *3 KSJ Research
Note: Status (same as Vol.2): data/sourced (gold) = DART or public source · judgment (shaded, brown) = unsourced setting, stress-tested · needed (red) = not set, fallback rule · verify = public basis, not yet checked · design = hypothetical setting · neutral = neutral shape without evidence

© 2026 Shinjae Kang · KSJ Research · Research Note Vol.3 · 2026.10 · For education and research only, not investment advice · Target Co. is hypothetical

Data notes: sources, dates and the SEC entity check

Series as-of 2026-10-20 · KRX 2026-10-19 · SEC 2026-10-20 · USD/KRW 1,332.1 (2026-10-20)

Data sources · datasets · dates

Source	Dataset / items	As of	Status
DART · FSS	Annual reports, consolidated, FY2022A–FY2025A (4 Korean names)	2026-10-20	Collected
KRX · Korea Exchange	Close, market cap, listed shares (2 Jan 2023 · 19 Oct 2026)	2026-10-20	Collected
SEC EDGAR	10-K annual financials (XBRL), cover shares, entity check	2026-10-20	Collected
yfinance (secondary)	U.S. peers' closes (first 2023 trading day · as-of)	2026-10-20	Collected
DART · SEC (extra)	D&A (for EBITDA) · SEC FY2022A NI	2026-10-20	Collected
ECOS · Bank of Korea	USD/KRW (display conversion)	2026-10-20	Collected

Note: Priority: official filings (DART, SEC) > public agencies (KRX, ECOS) > secondary libraries (yfinance, cross-checked with SEC shares)

SEC entity check, 5 names — 0 WARN; Vistra OK after checking the 10-K cover registrant [*1]

Company	Ticker	Listed CIK	Filer name (companyfacts)	Same group	Verdict
Vertiv	VRT	0001674101	Vertiv Holdings Co	yes	OK
Eaton	ETN	0001551182	EATON CORPORATION plc	yes	OK
GE Vernova	GEV	0001996810	GE Vernova Inc.	yes	OK
Constellation Energy	CEG	0001868275	CONSTELLATION ENERGY CORPORATION	yes	OK
Vistra	VST	0001692819	Vistra Operations Company LLC	yes (manual)	WARN → OK

Source: *1 SEC EDGAR 10-K XBRL (companyfacts), cover-page shares
Note: Market cap follows the listed entity (ticker map), financials the companyfacts filer — a mismatch drops the multiple (R3). Manual checks are recorded with URL and date; the automatic verdict is kept. Vistra source: https://www.sec.gov/Archives/edgar/data/1692819/000169281922000005/vistra-20211231.htm (checked 2026-10-22)

Methodology: one rule set for every peer and date; conclusions from the numbers

Every figure is calculated from data and assumptions, and conclusions are generated from the results

Data → multiples → price → checks and report [*1]

1. Data

DART · KRX

SEC EDGAR

yfinance (sec.)

Entity, share checks

2. Multiples

3-step screen

Agg., mean, median

Outliers R1–R5

3 scenarios

3. Price

Equity value

Post-IPO shares

Discount grid

Sensitivity

4. Checks, report

MC seed, convergence

Multiple, unit, FX checks

Conclusion titles

Korean & English

Built with an AI coding tool (Claude Code) for data links, multiples, Monte Carlo and report generation; question design, peer and assumption judgments and number checks by the author.

Source: *1 KSJ Research
Note: Full code and data sources: GitHub https://github.com/Cojae-Kang/ksj-ai-energy-asset-portfolio

© 2026 Shinjae Kang · KSJ Research · Research Note Vol.3 · 2026.10 · For education and research only, not investment advice · Target Co. is hypothetical

Glossary: the language of IPO pricing

Units: KRW 100mn and KRW, USD mn · %, x

Glossary

Term	Definition
PER	Price-to-earnings = market cap ÷ NI to parent; the main IPO metric
Aggregate P/E	Σcap ÷ ΣNI — an earnings-weighted average that damps high multiples on small profits
EV/EBITDA	Enterprise value (cap + net debt) ÷ EBITDA; a capital-structure-neutral cross-check
EV/EBIT	Proxy without D&A: EV ÷ operating income (higher than EV/EBITDA)
Value per share	Equity value ÷ post-IPO shares — the price before discount
IPO discount	1 − offer price ÷ value per share; compensation for post-listing uncertainty
Indicative price range	The price range offered before book-building (low–top)
Primary · secondary	Primary = new shares (cash to company); secondary = existing holders sell (cash to them)
Overhang	Existing holders' shares that may hit the market after lock-ups
Lock-up	A period after listing during which major holders cannot sell
Three-step peer screen	Industry → financial tests (profit, multiple, filer) → business fit
Cycle peak	Earnings and multiples both high; risk that slowing profits and de-rating coincide
Monte Carlo	Repeated draws from input distributions to get an output distribution (fixed seed = reproducible)
P10 · P50 · P90	10th, 50th and 90th percentiles of the outcome
Loss probability	P(1-yr value per share < offer price)

Note: Shaded = core IPO-pricing terms

© 2026 Shinjae Kang · KSJ Research · Research Note Vol.3 · 2026.10 · For education and research only, not investment advice · Target Co. is hypothetical

Disclosures

- This is a personal portfolio for education and research. It is not investment advice or a recommendation on any security.
- Target Co. is hypothetical and unrelated to any real company; the price range illustrates a method.
- Views are the author's own and do not represent any current or future employer.
- All data are public; accuracy and completeness are not guaranteed. Assumptions and sources are in Appendix A2.
- As of 2026-10-20. Methods and code can be reproduced from the repository.
- This report is a personal research portfolio prepared for IB and infrastructure-finance roles; its analytical logic and code are public. It is not affiliated with any financial institution or department.
- © 2026 Shinjae Kang. The report, charts and text are released under CC BY 4.0. Any citation or reuse must credit 'Shinjae Kang (KSJ Research)' and the source.
- Code is under the MIT License (copyright notice must be retained). Rights in cited third-party data and materials (IEA, EIA, CBRE, etc.) remain with their owners and are subject to their terms.

KSJ Research

https://github.com/Cojae-Kang/ksj-ai-energy-asset-portfolio

Contact: toonword@naver.com